# Defining Linear Functions

(Also available in Pyret)

Students explore the concept of slope and y-intercept in linear relationships, using function definitions as a third representation (alongside tables and graphs).

| | |
|---|---|
| **Lesson Goals** | Students will be able to…<br>• define linear functions using slope and y-intercept<br>• match different representations of the same linear relationship |
| **Student-facing Lesson Goals** | • I can define a function to express a linear relationship and identify the slope and y-intercept in a function definition.<br>• I can match different representations of the same linear relationship |
| **Prerequisites** | • Simple Data Types<br>• Contracts<br>• Functions Can Be Linear<br>• Functions: Contracts, Examples & Definitions |
| **Materials** | • Lesson Slides<br>• Identifying Slope and y-intercept in Definitions (Page 29)<br>• Matching Graphs to Function Definitions (Page 30)<br>• Summarizing Graphs with Function Definitions (Page 31)<br>• Matching Tables to Function Definitions (Page 32)<br>• Summarizing Tables with Function Definitions (Page 33)<br>• Matching Tables and Definitions (Desmos)<br>• Matching Graphs and Definitions of Linear Functions (Desmos) |
| **Key Points for the Facilitator** | • Lines are made of points (try to avoid referring to "lines" as much as "collections of points")<br>• Linear functions can be represented as straight lines on a graph or as sequences that change at a constant rate in a table.<br>• Stress that Graphs, Definitions, and Tables are all *different ways of viewing the same function*. The misconception to fight against is the idea that $f(x) = 3x + 2$ is somehow the "real" function, with the graph being "the thing the function draws." |

***Glossary***

**coordinate pair ::** a set of numbers describing an object's location on the coordinate plane

**domain ::** the type or set of inputs that a function expects

**range ::** the type or set of outputs that a function produces

**slope ::** the steepness of a straight line on a graph

**y-intercept ::** the point where a line or curve crosses the y-axis of a graph

# Defining Linear Functions *35 minutes*

## *Overview*

Students explore function definitions as a way of expressing linear relationships, and construct tables and graphs from those definitions.

## *Launch*

As you've seen, a **function definition** is a way of summarizing a relationship. You've seen how a linear relationship can be expressed as a table or graph. But what do these kinds of relationships look like as a definition?

Linear functions are defined by their ***slope*** and ***y-intercept***

The **slope-intercept form of the line** includes the slope as the coefficient of x and the y-intercept as the numerical term. You will hear people describe this form as $y = mx + b$, where $m$ stands for slope and $b$ stands for the y-intercept.
While it is common to write the x-term first and the y-intercept second, they can be written in any order!

| Function Notation | WeScheme Code |
|---|---|
| $f(x) = 6x - 10$ | `(define (f x) (- (* 6 x) 10))` |
| $f(x) = -10 + 6x$ | `(define (f x) (+ -10 (* 6 x)))` |

When the slope is zero (and the line is horizontal)... we may choose whether or not to write the slope term.

| "Visible" Slope | "Invisible" Slope |
|---|---|
| $f(x) = 0x + 22$ | $f(x) = 22$ |

When the y-intercept is $0$ (and the line crosses the y-axis at the origin)... we may choose whether or not to write the slope term.

| "Visible" y-intercept | "Invisible" y-intercept |
|---|---|
| $f(x) = 3.2x + 0$ | $f(x) = 3.2x$ |

To check our work, we can apply the function to the x-value from any ***coordinate pair*** on our table or graph, and it should produce the y-value!
As with tables and graphs, a function definition can also reveal whether or not the function is linear. Functions that are not linear will follow other forms, for example they may include exponents or absolute values.

## *Investigate*

**Let's start by identifying the slope and y-intercept from function definitions.**

- Identifying Slope and y-intercept in Definitions

*More practice is available through our Desmos card sort activities:*

- Identifying y-intercepts in Tables, Graphs & Definitions of Linear Functions (Desmos)
- Identifying slope in Tables, Graphs & Definitions of Linear Functions (Desmos)

**Let's connect definitions to tables and graphs.**
Writing down the slope and y-intercept beneath each representation will help!

- Matching Tables to Function Definitions
- Matching Graphs to Function Definitions

*More matching practice is available through our Desmos card sort activities.*

- Matching Tables and Definitions (Desmos)
- Matching Graphs and Definitions of Linear Functions (Desmos)
- Matching Graphs and Definitions of Linear Functions 2 (Desmos)

**Let's write our own definitions from tables and graphs!**

- Summarizing Tables with Function Definitions
- Summarizing Graphs with Function Definitions

What strategies did you use?

## *Common Misconceptions*

It is common to think of the graph as the "output" of the function, rather than the function itself. Most math textbooks will use language like "matching the graph to the function", suggesting that the graph is somehow not the function! Since this language is pervasive, it's important to actively push against it.

## *Synthesize*

Function definitions are a way of talking about relationships between quantities: milk costs $0.59/gallon, a stone falls at $9.8m/s^2$, or there are 30 students for every teacher at a school. If we can figure out the relationship between a small sample of data, we can *make predictions* about what happens next. We can see these relationships as tables, graphs, or symbols in a definition. We can even think about them as a mapping between ***Domain*** and ***Range***!

When we talk about functions, it's helpful to be able to switch between representations, and see the connections between them.

# Finding the y-intercept from the Slope and a Point *20 minutes*

## *Launch*

Consider the function $f(x) = 3x$.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y | 0 | 3 | 6 | 9 |

- What is the slope?
  - *3*
- What is the y-intercept?
  - *0*
- What is the y-value when x = 2?
  - *6*

Anytime the y-intercept is 0, we can multiply any x-value by the slope to get its corresponding y-value.

But if the y-intercept isn't zero… there is another step to finding the y-value.

Consider the function $f(x) = 3x - 2$.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y | -2 | 1 | 4 | 7 |

- What is the slope?
  - *3. Same as for the previous function.*
- What is the y-intercept?
  - *-2*
- What is the y-value when x = 2?
  - *4. Two less than the y-value for $x = 3$ in the previous function, where the y-intercept was 0.*

The y-intercept always gets added to / subtracted from the product of the slope and the x-value to find the corresponding y-value.

## *Investigate*

As discussed above, the relationship between the x-values and the y-values can be described using $y = mx + b$, where $m$ stands for ***slope*** and $b$ stands for the ***y-intercept***.

If we solve that for the y-intercept…

$b = y - mx$

In other words, *the y-intercept can be calculated by subtracting the product of the slope and any x-value from the corrseponding y-value.*

Let's say the slope is $3$. And we know that the line passes through the point (7,9).

- $b = y - mx$
- $m = 3$
- $x = 7$
- $y = 9$
- so... $b = 9 - 3(7)$

To find the y-intercept, subtract $9$ *(the y-value of the point)* minus $3 \times 7$ *(the product of the slope and the x-value of the point).*

$$9 - 21 = -12$$

*y-intercept:* $-12$

*function definition making use of the y-intercept we found:* $f(x) = 3x - 12$

Consider the table below.

| x | 80 | 81 | 82 | 83 |
|---|---|---|---|---|
| y | 150 | 155 | 160 | 165 |

- What is the slope?
  - *5*
- Calculate the y-intercept using the first coordinate pair.
  - *-250*
- Do you get the same y-intercept if you use another pair?
  - *Yes.*

## Additional Exercises:

- Matching Function Definitions, Tables and Graphs (Desmos)
- Matching Function Definitions, Tables and Graphs 2 (Desmos)
- Matching Graphs & Definitions of Functions (not just linear!) (Desmos)
- Identifying y-intercepts in Tables, Graphs & Definitions of Linear Functions (challenge) (Desmos)
- Identifying Linearity in Tables, Graphs & Definitions of Linear Functions (Desmos)
- Matching Tables, Graphs, and Definitions of Functions (challenge!) (Desmos)
- Matching Tables, Graphs, and Definitions of Functions (challenge!) (Desmos)
- Identifying y-intercepts in Tables, Graphs & Definitions of Linear Functions (challenge!) (Desmos)